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Inventor(s)	Dunn; David Scott et al.

Laundry appliance drainage

Abstract

A laundry appliance includes a cabinet defining an interior volume with a housing mounted within the interior volume of the cabinet. An air filter is positioned in the housing. The air filter may be positioned upstream of a heating system of the laundry appliance. The laundry appliance also includes a drain path extending through the housing.

Inventors: Dunn; David Scott (Louisville, KY), Djerekarov; Jivko Ognianov (Louisville, KY), Lawson; Wayne E. (La Grange, KY), Tcholakov; Dimitar (Louisville, KY)

Applicant: Haier US Appliance Solutions, Inc. (Wilmington, DE)

Family ID: 1000008780269

Assignee: Haier US Appliance Solutions, Inc. (Wilmington, DE)

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10087569	12/2017	Bocchino	N/A	N/A
10619287	12/2019	Lu et al.	N/A	N/A
10633785	12/2019	Bocchino	N/A	N/A
10662594	12/2019	Lafoy	N/A	N/A
2017/0350063	12/2016	Bombardieri	N/A	D06F 58/22
2020/0399820	12/2019	Bombardieri	N/A	N/A
2021/0156070	12/2020	Rios Acebal	N/A	D06F 39/10

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2436832	12/2015	EP	N/A
2719819	12/2016	EP	N/A
WO2016095968	12/2015	WO	N/A
WO2022022170	12/2021	WO	N/A
WO2022142252	12/2021	WO	N/A

Primary Examiner: Ko; Jason Y

Attorney, Agent or Firm: Dority & Manning, P.A.

Background/Summary

FIELD OF THE INVENTION

(1) The present subject matter relates generally to laundry appliances, and more particularly to drainage features within laundry appliances.

BACKGROUND OF THE INVENTION

(2) Laundry appliances include washing machine appliances, dryer appliances, and combination laundry appliances. Combination laundry appliances, sometimes also referred to as washer/dryer appliances, provide both washing and drying functions in a single unit. During the washing and/or drying operations, particles from clothing articles being treated therein, such as fiber particles, are dislodged and may become entrained in the flow of air through the laundry appliance. Such particles, e.g., lint, may impair the performance of the laundry appliance, such as a heating system thereof, if the particles are carried to, for example, a heater of the heating system, by the flow of air. Thus, laundry appliances, e.g., dryer appliances and combination appliances, typically include at least one air filter, sometimes also referred to as a lint filter, to entrap such particles and remove the particles from the air flow.

(3) Laundry appliances, such as combination appliances, may contain various liquids, such as wash liquid, at various points during the operations thereof. These various liquids are generally removed from the laundry appliance by draining to a sump and pumping out from the sump. However, some

such liquids may become entrapped within the laundry appliance such that the liquid accumulates in an undesired area where a volume of the liquid is prevented or obstructed from reaching the sump. For example, in some instances, liquid may become entrapped at or around the air filter or lint filter. The entrapped liquid may lead to issues such as poor consumer perception when the filter is accessed for cleaning, such as when liquid entrapped behind or around the filter escapes as the consumer accesses the filter.

(4) Accordingly, a laundry appliance having improved features for drainage, e.g., around the filter, such as in a filter box or housing of the laundry appliance, would be advantageous.

BRIEF DESCRIPTION OF THE INVENTION

(5) Aspects and advantages of the invention will be set forth in part in the following description, or may be obvious from the description, or may be learned through practice of the invention.

(6) In one aspect of the present disclosure, a laundry appliance is provided. The laundry appliance includes a cabinet that defines an interior volume with a tub mounted within the interior volume of the cabinet. A laundry basket is rotatably mounted within the tub. The laundry basket defines a chamber for the receipt of articles for treatment. The laundry appliance also includes a heating system in thermal communication with the chamber whereby heated air flows from the heating system to the chamber. An air filter is positioned between the laundry basket and the heating system upstream of the heating system, whereby a flow of return air from the chamber passes through and is filtered by the air filter before flowing to the heating system. The laundry appliance also includes a housing. The air filter is positioned within the housing and a drain path extends through the housing.

(7) In another aspect of the present disclosure, a laundry appliance is provided. The laundry appliance includes a cabinet that defines an interior volume with a housing mounted within the interior volume of the cabinet. An air filter is positioned in the housing. The laundry appliance also includes a drain path extending through the housing.

(8) These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

(2) FIG. 1 provides a perspective view of a laundry appliance in accordance with one or more exemplary embodiments of the present disclosure.

(3) FIG. 2 provides a schematic cross-section view of the example laundry appliance of FIG. 1.

(4) FIG. 3 provides a schematic diagram of an exemplary heat exchange heating system of the example laundry appliance of FIG. 1 according to one or more embodiments of the present disclosure.

(5) FIG. 4 provides a perspective view of the laundry appliance of FIG. 1 with a portion of a cabinet thereof removed to show internal components of the laundry appliance in accordance with one or more exemplary embodiments of the present disclosure.

(6) FIG. 5 provides another perspective view of the laundry appliance of FIG. 1 with an upper portion of a housing removed to show an air filter assembly and a heating system therein in accordance with one or more exemplary embodiments of the present disclosure.

(7) FIG. 6 provides a perspective view of a portion of the housing, the air filter assembly, and the heating system of FIG. 5.

(8) FIG. 7 provides a perspective view of the filter assembly of FIG. 5.

(9) FIG. 8 provides another perspective view of a portion of the housing of FIG. 5.

(10) FIG. 9 provides another perspective view of a portion of the housing of FIG. 5.

(11) FIG. 10 provides an enlarged perspective view of a portion of the housing of FIG. 5.

(12) FIG. 11 provides a partially sectioned enlarged perspective view of a portion of the housing and the air filter assembly.

DETAILED DESCRIPTION

(13) Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope or spirit of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

(14) As used herein, terms of approximation, such as “substantially,” “generally,” or “about” include values within ten percent greater or less than the stated value. When used in the context of an angle or direction, such terms include within ten degrees greater or less than the stated angle or direction. For example, “generally vertical” includes directions within ten degrees of vertical in any direction, e.g., clockwise or counter-clockwise.

(15) As used herein, the terms “articles,” “clothing,” or “laundry” include but need not be limited to fabrics, textiles, garments, linens, papers, or other items which may be cleaned, dried, and/or otherwise treated in a laundry appliance. Furthermore, the term “load” or “laundry load” refers to the combination of clothing that may be washed together in a washing machine appliance or dried together in a dryer appliance (e.g., clothes dryer), including washed and dried together in a combination laundry appliance, and may include a mixture of different or similar articles of clothing of different or similar types and kinds of fabrics, textiles, garments and linens within a particular laundering process.

(16) Embodiments of the present disclosure include a laundry appliance, e.g., a combination washer-dryer appliance, such as the example combination appliance **10** illustrated in FIGS. **1** through **5**. FIG. **1** provides a perspective view of a laundry appliance **10** according to exemplary embodiments of the present disclosure. The laundry appliance **10** is a combination laundry appliance, and may also be referred to as a multifunction laundry appliance or washer/dryer combination appliance. FIG. **2** provides a section view of laundry appliance **10**. FIG. **3** provides a schematic illustration of a heat pump heating system which may be incorporated into the laundry appliance **10**. FIGS. **4** and **5** provide additional perspective views of the laundry appliance **10**. The laundry appliance **10** generally defines a vertical direction V, a lateral direction L, and a transverse direction T, each of which is mutually perpendicular, such that an orthogonal coordinate system is defined. While described in the context of a specific embodiment of laundry appliance **10**, using the teachings disclosed herein, it will be understood that laundry appliance **10** is provided by way of example only. Other laundry appliances having different appearances and different features may also be utilized with the present subject matter as well.

(17) Cabinet **12** includes a front panel **14**, a rear panel **16**, a left side panel **18** and a right side panel **20** spaced apart from each other by front and rear panels **14** and **16**, a bottom panel **22**, and a top cover **24**. As used herein, terms such as “left” and “right” or “front” and “back” refer to directions from the perspective of a user facing the laundry appliance **10** for accessing and/or operating the laundry appliance **10**. For example, a user stands in front of the laundry appliance **10**, e.g., at or near the front panel **14**, to access door **33** and/or inputs **70** (the door **33** and inputs **70** are described

in more detail below). Within cabinet **12**, an interior volume **29** is defined. A drum or tub **26** is mounted within the interior volume **29**. A laundry basket **130** is mounted within the tub **26**. The laundry basket **130** defines a chamber **25** for receipt of articles of clothing for treatment, e.g., washing, rinsing, spinning, tumbling, and/or drying.

(18) In some embodiments, one or more selector inputs **70**, such as knobs, buttons, touchscreen interfaces, etc., may be provided or mounted on the cabinet **12**, e.g., on a control panel **71** thereof and are in operable communication (e.g., electrically coupled or coupled through a wireless network band) with a processing device or controller **56**. The control panel **71** may also include a display **64**. Controller **56** may also be provided in operable communication with various components of the laundry appliance, such as the motor, blower, and/or heating system **80**. In turn, signals generated in controller **56** direct operation of such components in response to the position of inputs **70**. As used herein, “processing device” or “controller” may refer to one or more microprocessors, microcontroller, ASICs, or semiconductor devices and is not restricted necessarily to a single element. The controller **56** may be programmed to operate laundry appliance **10** by executing instructions stored in memory (e.g., non-transitory media). The controller **56** may include, or be associated with, one or more memory elements such as RAM, ROM, or electrically erasable, programmable read only memory (EEPROM). For example, the instructions may be software or any set of instructions that when executed by the processing device, cause the processing device to perform operations. It should be noted that controllers as disclosed herein are capable of and may be operable to perform any methods and associated method steps as disclosed herein. For example, in some embodiments, methods disclosed herein may be embodied in programming instructions stored in the memory and executed by the controller.

(19) Tub **26** extends between a front portion **37** and a back portion **38**. Tub **26** is generally cylindrical in shape, having an outer cylindrical wall **28** and a front flange or wall **30** that defines an opening **32** of tub **26**, e.g., at front portion **37** of tub **26**, for loading and unloading of articles into and out of a chamber **25** defined by and within a laundry basket **130** inside of the tub **26**. Tub **26** includes a rear wall **34** opposite the front flange **30**. A door **33** provides for closing or accessing tub **26** through opening **32**. A window **36** (FIG. **1**) may be provided in door **33** for viewing of the chamber **25** and/or laundry articles therein, e.g., during operation of the laundry appliance **10**.

(20) Laundry basket **130** is rotatably mounted within tub **26** such that the laundry basket **130** is rotatable about an axis of rotation. According to the illustrated embodiment, the axis of rotation is substantially parallel to the transverse direction T. In this regard, laundry appliance **10** is generally referred to as a “horizontal axis” or “front load” laundry appliance **10**. However, it should be appreciated that aspects of the present subject matter may be used within the context of a vertical axis or top load laundry appliance as well.

(21) Laundry appliance **10** includes a motor assembly **126** that is in mechanical communication with laundry basket **130** to selectively rotate laundry basket **130**. The motor assembly **126** may be a pancake motor, as illustrated, or any other suitable type, size, or configuration of motor may be used to rotate laundry basket **130** according to various embodiments. For example, a motor, such as a brushless DC motor, may be mounted within the cabinet **12** and the motor may be coupled to the laundry basket **130** by a belt and pulley, whereby the motor rotates the laundry basket **130** through the belt and pulley.

(22) Laundry basket **130** may define one or more agitator features that extend into chamber **25** to assist in agitation and cleaning of articles disposed within laundry chamber **25** during operation of laundry appliance **10**. For example, as illustrated in FIG. **2**, a plurality of ribs **128** extends from laundry basket **130** into chamber **25**. In this manner, for example, ribs **128** may lift articles disposed in laundry basket **130** during rotation of laundry basket **130**, such as during an agitation or rinse portion of a wash operation of the laundry appliance **10**. During a drying operation of the laundry appliance **10**, the ribs **128** may also lift articles in the chamber **25** of the laundry basket **130** and then allow such articles to tumble back to a bottom of laundry basket **130** as laundry basket **130**

rotates.

(23) As illustrated for example in FIG. 2, laundry basket **130** may also include a plurality of perforations **140** extending therethrough in order to facilitate fluid communication between chamber **25** and tub **26**, e.g., whereby wash liquid may flow between the tub **26** and the chamber **25** during a wash operation or cycle and/or heated air may flow into the chamber **25** and moisture-laden air may flow out of the chamber **25** during a drying operation or cycle. A sump **142** is defined by tub **26** outside of laundry basket **130** at a bottom of the tub **26** along the vertical direction V. Thus, sump **142** is configured for receipt of, and generally collects, wash liquid (the wash liquid may include, e.g., water, and may also includes additives such as detergents, etc.) during wash operations of laundry appliance **10**. For example, during a wash operation of laundry appliance **10**, wash liquid may be urged (e.g., by gravity) from the chamber **25** within the laundry basket **130** to sump **142** through the plurality of perforations **140**. A pump assembly **40** is located beneath tub **26** for gravity assisted flow when draining tub **26** (e.g., via a drain **41**). Pump assembly **40** is also configured for recirculating wash liquid within tub **26**.

(24) In some embodiments, laundry appliance **10** includes an additive dispenser or spout **150**. For example, spout **150** may be in fluid communication with a water supply (not shown) in order to direct fluid (e.g., clean water) into tub **26**. Spout **150** may also be in fluid communication with the sump **142**. For example, pump assembly **40** may direct wash liquid disposed in sump **142** to spout **150** in order to circulate wash liquid in tub **26**.

(25) As illustrated, a detergent dispenser drawer **152** may be slidably mounted within front panel **14**. Detergent dispenser drawer **152** receives an additive (e.g., detergent, fabric softener, bleach, or any other suitable liquid or powder) and directs the additive to chamber **25** during operation of laundry appliance **10**. According to the illustrated embodiment, detergent dispenser drawer **152** may also be fluidly coupled to spout **150** to facilitate the complete and accurate dispensing of the additive.

(26) In exemplary embodiments, during operation of laundry appliance **10**, laundry items are loaded into laundry basket **130** through opening **32**, and an operation is initiated through operator manipulation of input selectors **70**. For example, a wash cycle may be initiated such that tub **26** is filled with water, detergent, or other fluid additives (e.g., via spout **150**). One or more water valves (not shown) can be controlled by laundry appliance **10** to provide for filling laundry basket **130** to the appropriate level for the amount of articles being washed or rinsed. By way of example, once laundry basket **130** is properly filled with fluid, the contents of laundry basket **130** can be agitated (e.g., with ribs **128**) for an agitation phase of laundry items in laundry basket **130**. During the agitation phase, the basket **130** may be motivated about the axis of rotation at a set speed (e.g., a tumble speed) by a motor. As the basket **130** is rotated, articles within the basket **130** may be lifted by ribs **128** and permitted to drop therein due to gravity.

(27) After the agitation phase of the washing operation is completed, tub **26** can be drained. Laundry articles can then be rinsed (e.g., through a rinse cycle) by again adding fluid to tub **26**, depending on the particulars of the cleaning cycle selected by a user. Ribs **128** may again provide agitation within laundry basket **130**. One or more spin cycles may also be used. In particular, a spin cycle may be applied after the wash cycle or after the rinse cycle in order to wring wash liquid from the articles being washed. During a spin cycle, basket **130** is rotated at relatively high speeds. For instance, basket **130** may be rotated at one set speed (e.g., a pre-plaster speed) before being rotated at another set speed (e.g., a plaster speed). As would be understood by those of ordinary skill in the art, the pre-plaster speed may be greater than the tumble speed and the plaster speed may be greater than the pre-plaster speed. Moreover, agitation or tumbling of articles may be reduced as basket **130** increases its rotational velocity such that the plaster speed maintains the articles at a generally fixed position relative to basket **130**.

(28) After the spin cycle, a drying operation may begin. A supply duct **82** may be mounted to tub **26** and may extend between tub **26** and a heating assembly or system **80**, whereby the tub **26** is

downstream of the heating system **80** along the supply duct **82** such that the heating system **80** supplies heated air that has been heated by the heating system **80** to the tub **26** via the supply duct **82**. A return duct **84** may also be mounted to the tub **26** and may extend between tub **26** and the heating system **80** whereby the tub **26** is upstream of the heating system **80** along the return duct **84** such that the heating system **80** receives relatively warm and humid air from the tub **26** via the return duct **84**, e.g., air returns to the heating system **80** from the tub **26**, e.g., after flowing over and around articles within the chamber **25**, through the return duct **84**. In some embodiments, e.g., as illustrated in FIGS. **2** and **3**, the heating system **80** may be positioned above the tub **26** and the return duct **84** may be oriented generally along the vertical direction **V**, and the return duct **84** may also be referred to as a chimney. The supply duct **82** may be mounted to the tub **26**, e.g., at the rear wall **34** thereof as in the illustrated example, or the supply duct **82** may be mounted to the cylindrical wall **28** of the tub **26**, such as above the basket **130**, similar to the return duct **84** in the illustrated example embodiment.

(29) The heating system **80**, may include, e.g., a resistance heating element, a gas burner, and/or a heat pump, such as the example heat pump embodiment illustrated in FIG. **3** and described in more detail below, or any other suitable heat source. Moisture-laden, warm air is drawn from tub **26** by an air handler, such as a blower fan, which generates a negative air pressure within the chamber **25**. As the air passes from the blower fan, it enters return duct **84** and then is passed into heating system **80**. Heated air (with a lower moisture content than was received from tub **26**), exits heating system **80** and is supplied to tub **26** by supply duct **82**. After the clothing articles have been dried, they are removed from the chamber **25** via opening **32**.

(30) Turning now to FIG. **3**, a schematic view of selected components of one or more exemplary embodiments of laundry appliance **10** is provided. In particular, FIG. **3** illustrates components used during drying operations of the laundry appliance **10**. It is understood that, except as otherwise indicated, laundry appliance **10** in FIG. **3** may include some or all of the features described herein with respect to FIGS. **1**, **2**, **4**, and **5**.

(31) In operation, one or more laundry articles **1000** may be placed within the chamber **25** of laundry basket **130**. Hot dry air **118** may be supplied to chamber **25** whereby moisture within laundry articles **1000** may be drawn from the laundry articles **1000** by evaporation, such that warm saturated air **120** may flow from chamber **25** to an evaporator **102** of the heating system **80**, e.g., via the return duct **84** illustrated in FIG. **2**. As air passes across evaporator **102**, the temperature of the air is reduced through heat exchange with refrigerant that is vaporized within, for example, coils or tubing of evaporator **102**. This vaporization process absorbs both the sensible and the latent heat from the moisture-laden air-thereby reducing its temperature. As a result, moisture in the air is condensed and such condensate may be drained from heating assembly **80**, as will be understood by those of ordinary skill in the art.

(32) Air passing over evaporator **102** becomes drier and cooler than when it was received from tub **26** of laundry appliance **10**. As shown, cool dry air **122** from evaporator **102** is subsequently caused to flow across a condenser **108** (e.g., across coils or tubing of the condenser **108**), which condenses refrigerant therein. The refrigerant enters condenser **108** in a gaseous state at a relatively high temperature and pressure compared to the cool, dry air **122** from evaporator **102**. As a result, heat energy is transferred to the air at the condenser section **108**, thereby elevating the temperature of the air and providing hot dry air **118** for supply to the tub **26** of laundry appliance **10**, e.g., via the supply duct **82** illustrated in FIG. **2**. The hot dry air **118** passes over and around laundry articles **1000** within the chamber **25** inside of the tub **26**, such that warm saturated air **120** is generated, as mentioned above. For example, the hot dry air may circulate around and through the articles **1000** while the articles **1000** are tumbled within the chamber **25** such as by rotating the basket **130**, and the tumbling may be promoted by ribs **128** as well. Because the air is recycled through tub **26** and heating system **80**, e.g., in a closed-loop as mentioned above, laundry appliance **10** can have a much greater efficiency than traditional clothes dryers where warm, moisture-laden air is exhausted

to the environment.

(33) As shown in FIG. 3, some embodiments of heating system **80** include a compressor **104** that pressurizes refrigerant (i.e., increases the pressure of the refrigerant) supplied by a suction line **110** and generally motivates refrigerant through the sealed refrigerant circuit of heating system **80**. Compressor **104** may be in operable communication with controller **56** and is generally designed to pressurize a gas phase refrigerant. Accordingly, in order to avoid damage, refrigerant in suction line **110** is supplied to the compressor **104** in a gas phase from the evaporator section **102**. The pressurization of the refrigerant with compressor **104** increases the temperature of the refrigerant (e.g., as directed by controller **56**). The compressed refrigerant is fed from compressor **104** to condenser **108** through line **112**. As relatively cool air **122** from the evaporator **102** is passed over the condenser **108**, the refrigerant is cooled and its temperature is lowered as heat is transferred to the air for supply to tub **26**.

(34) Upon exiting condenser **108**, the refrigerant is fed through line **114** to an expansion device **106**. Although only one expansion device **106** is shown, such is by way of example only. It is understood that multiple such devices may be used. In the illustrated example, expansion device **106** is a thermal expansion valve. In additional embodiments, any other suitable expansion device, such as a capillary tube, may be used as well as or instead of the thermal expansion valve **106**. Expansion device **106** lowers the pressure of the refrigerant and controls the amount of refrigerant that is allowed to enter the evaporator **102** via line **116**. Importantly, the flow of liquid refrigerant into evaporator **102** is limited by expansion device **106** in order to keep the pressure low and allow expansion of the refrigerant back into the gas phase in the evaporator **102**. The evaporation of the refrigerant in the evaporator **102** converts the refrigerant from its liquid-dominated phase to a gas phase while cooling and drying the air **120** from tub **26**. The process is repeated as air is circulated through tub **26** and between evaporator **102** and condenser **108** while the refrigerant is cycled through the sealed refrigerant circuit, as described above.

(35) The laundry appliance **10** may include an air filter **202**, e.g., positioned between the tub **26** and the heating system **80** with the air filter **202** downstream of the tub **26**, e.g., downstream of the chamber **25** of the basket **130** in the tub **26**, and upstream of the heating system **80**, whereby air flowing from the chamber **25** passes through and is filtered by the air filter **202** before flowing to the heating system **80**.

(36) As mentioned, the air filter **202** filters the air, e.g., the air filter **202** may remove particles such as lint particles from the air, as well as dust, hair, or other airborne matter, as will be understood by those of ordinary skill in the art. Accordingly, such particles may accumulate on and in the air filter **202**, and in particular on an outer surface or upstream surface of the air filter **202**, over time. If left unchecked, such accumulated particles may eventually obstruct air flow through the filter **202**, resulting in reduced efficiency or performance of the air filter **202** and/or the laundry appliance **10**, such as the heating system **80** of the laundry appliance **80**. Thus, the air filter **202** may be user accessible, e.g., to permit a user to clean the air filter **202**.

(37) FIG. 4 provides a perspective view of laundry appliance **10** with a portion, e.g., an upper portion such as top cover **24**, of the cabinet **12** thereof removed to reveal internal components of the laundry appliance **10**. The laundry appliance **10** may include a heating system **80**, e.g., as described above. As illustrated in FIG. 4, the heating system **80** is enclosed within a housing **200**. The heating system **80** may be seen in FIG. 5, where an upper portion, e.g., upper half, of the housing **200** is also removed, e.g., in addition to the portion of the cabinet **12**. The heating system **80** may include a heat exchanger, e.g., evaporator **102** (see, e.g., FIG. 3) and an air filter **202** may be positioned between the laundry basket **130** and the heating system **80**. The air filter **202** may be a part of an air filter assembly **201**. For example, the air filter **202** may be positioned upstream of the heating system **80**, such as upstream of the evaporator **102** of the heating system **80**, e.g., where the evaporator **102** is the furthest upstream portion or component of the heating system **80**, e.g., the first component of the heating system **80** to receive a flow of warm, moist air from the chamber **25**.

With the air filter **202** so positioned, a flow of return air **120** (FIG. 3) from the chamber **25** may pass through and may be filtered by the air filter **202** before flowing to the heating system **80**. For example, the air filter **202** may include a porous or air-permeable material which permits air to flow therethrough while trapping particles above a specified size therein. The air filter **202** and/or air filter assembly **201** may be removable from the housing **200**.

(38) FIG. 6 provides a perspective view of a lower portion, e.g., lower half, of the housing **200** (e.g., the upper portion of the housing is removed in FIG. 6 in order to depict internal structures of the housing **200** and to depict the air filter assembly **201** in the inserted position in the housing **200**). As illustrated in FIG. 6, the air filter **202** may be part of the air filter assembly **201**. The air filter assembly **201** may include an outer shell **208** which interfaces with the housing **200** and a frame **206** mounted in the shell **208**. The air filter assembly **201** may also include a handle **222** (FIG. 5), e.g., on an outer end of the shell **208**, such as may be grasped by a user to extract the air filter assembly **201** from the housing **200**. The air filter **202** (FIG. 11) may be mounted to the frame **206**. For example, the frame **206** may be releasably mounted in the shell **208** and/or the filter **202** may be releasably mounted to the frame **206**, such as to remove the filter **202** and/or frame **206** for cleaning, e.g., to remove accumulated dust, lint, and/or other similar particles which may become entrapped in or on the air filter **202** over time after use. For example, the air filter **202** may be removed from the air filter assembly **201**, e.g., after taking the air filter assembly **201** out of the housing **200**.

(39) As may be seen in FIG. 6, the housing **200** may include a floor **204**. The floor **204** may be pitched or sloped, such as pitched towards an outlet **212** of the return duct **84**. The pitch of the floor **204** may define or partially define a drain path **210** that extends through the housing **200**, such as at least partially through the housing **200**, such as from downstream of the air filter **202** and/or air filter assembly **201** to the outlet **212** of the return duct **84**, where “downstream” of the air filter assembly **201** refers to the flow direction of air returning to the heating system **80** from the chamber **25**, e.g., such that the drain path **210** for liquid from the housing **200** may be generally opposite the direction of air flow. Thus, for example, moisture which enters the housing **200** or which otherwise accumulates within the housing **200**, such as accumulates downstream of the air filter assembly **201**, may be directed away from the heating assembly **80**, e.g., to or towards the return duct **84**, such as to permit the moisture to flow to the sump **142** (FIG. 2).

(40) Also as may be seen in FIGS. 6 and 7, the air filter **202** and/or air filter assembly **201** may include an upstream side **240** and a downstream side **242**, e.g., with respect to the flow of return air **120** from the chamber **25** to the heating assembly **80**. For example, the downstream side **242** of the air filter **202** and/or air filter assembly **201** may be proximate to and face towards the heating system **80**, whereas the upstream side **240** may be opposite the downstream side **242**, e.g., the upstream side may be the farthest side of the air filter assembly **201** from the heating system **80** and may face away from the heating system **80**, such as towards the outlet **212** of the return duct **84**. In such embodiments, the drain path **210** may extend between the upstream side **240** of the air filter **202** and the downstream side **242** of the air filter **202**, such as from the downstream side **242** to the upstream side **240** and/or to other components upstream of the air filter **202**, e.g., to the return duct **84** and/or the outlet **212** thereof.

(41) FIG. 7 provides a perspective view of an exemplary air filter assembly **201** according to one or more embodiments of the present disclosure. As illustrated in FIG. 7, the air filter assembly **201** may include an outer shell **208** which interfaces with the housing **200** and a frame **206** mounted in the shell **208**. The air filter assembly **201** may also include a handle **222** (FIGS. 4 and 5), e.g., on an outer end of the shell **208**, such as may be grasped by a user to extract the air filter assembly **201** from the housing **200**. The air filter **202** may be mounted to the frame **206**. For example, the frame **206** may be releasably mounted in the shell **208** and/or the filter **202** may be releasably mounted to the frame **206**, such as to remove the filter **202** and/or frame **206** for cleaning, e.g., to remove accumulated dust, lint, and/or other similar particles which may become entrapped in or on the air

filter **202** over time after use. For example, the air filter **202** may be removed from the air filter assembly **201**, e.g., after taking the air filter assembly **201** out of the housing **200**.

(42) In some embodiments, e.g., as illustrated in FIG. 7, the air filter assembly **201** may also include a latch **264**. For example, the latch **264** may be mounted on, such as integrally joined to, a cantilevered arm **266**. In some embodiments, the cantilevered arm **266** may be joined to the shell **208** of the air filter assembly **201**.

(43) Also as may be seen in FIG. 7, the air filter assembly **201** may further include a gasket **224**, e.g., on the shell **208** and surrounding the air filter **202**. The air filter **202** and/or air filter assembly **210** may include an upstream side **240** and a downstream side **242**, e.g., with respect to the flow of return air from the chamber **25** to the heating assembly **80**. For example, the downstream side **242** of the air filter **202** and/or air filter assembly **201** may be proximate to and face towards the heating system **80**, whereas the upstream side may be opposite the downstream side **242**, e.g., the upstream side may be the farthest side of the air filter assembly **201** from the heating system **80** and may face away from the heating system **80**, such as towards the return duct **84**. For example, the gasket **224** may be disposed on the downstream side **242** of the air filter assembly **201**, e.g., whereby the gasket **224** sealingly engages the heating system **80** when the air filter assembly **201** is in an installed position within the housing **200**.

(44) In some embodiments, such as the example embodiments illustrated in FIGS. **8** and **9**, the drain path **210** may be a divided drain path, whereby accumulated moisture may travel across one or more branches of the drain path **210** before ultimately ending up at the end point of the drain path **210**, e.g., at the outlet **212** of the return duct **84** from which air flows into the housing **200** (the terms “outlet” and “into” are used with respect to the air flow direction, as mentioned above with respect to “downstream” of the air filter). For example, as may be seen in FIGS. **8** and **9**, the housing **200** may include a guide rail **214** and the guide rail **214** may be discontinuous, e.g., one or more gaps **216** may be formed in the guide rail **214**. Where more than one gap **216** is provided in the guide rail **214**, each gap **216** may permit moisture to flow therethrough, thus the gap **216** or each gap **216** may define a portion of the drain path **210**, such as a branch of the drain path **210** in embodiments where the drain path **210** is a divided drain path.

(45) As mentioned, the air filter **202** and/or air filter assembly **201** may be removably positioned in the housing **200**. In such embodiments, the air filter **202** and/or air filter assembly **201** may be configured to translate in a travel direction along the guide rail **214** between an inserted position in the housing **200** (see, e.g., FIGS. **4** through **6**), and an extracted position at least partially outside of the housing **200**, such as where the air filter **202** and/or air filter assembly **201** may be entirely outside of the housing **200**, e.g., as illustrated in FIGS. **8** through **10**. For example, the housing **200** may include a filter gate **218** through which the air filter assembly **201** may be inserted into and/or removed from the housing **200**. In embodiments where the guide rail **214** is discontinuous, the guide rail **214** may be discontinuous along the travel direction, e.g., the guide rail **214** may be oriented generally along the travel direction and the discontinuities, e.g., gaps **216**, may also be oriented generally along and/or generally parallel to the travel direction.

(46) In some embodiments, e.g., as illustrated in FIGS. **8** through **10**, the housing **200** may include a water channel **220**, e.g., defined in the floor **204** of the housing **200**. In such embodiments, the water channel **220** may define at least a portion of the drain path **210**. For example, in some embodiments, e.g., as may be seen in FIGS. **8** through **10**, the water channel **220** may coincide with or be aligned with the gap **216** in the guide rail **214**, or one of the gaps **216** in embodiments where multiple gaps **216** are included. In particular, as may be seen in FIGS. **8** through **10**, the water channel **220** may be coincident with a gap **216** proximate to the gate **218**, such as the gap **216** of the plurality of gaps **216** which is most proximate to the gate **218**, e.g., in embodiments where multiple gaps **216** are included. In such embodiments, moisture which is retained in or around, e.g., under, the air filter assembly **201** may be released or permitted to escape via the water channel **220** as the air filter assembly **201** is removed from the housing **200**, whereupon the moisture may then

continue along the drain path **210**, e.g., from the water channel **220** to the outlet **212** of the return duct **84**.

(47) As may be seen in FIG. **11**, the air filter assembly **201** includes a guide surface **230**, and the guide surface **230** may be oriented downwards along the vertical direction V. When the air filter is in the inserted position and/or a partially inserted position, the guide surface **230** may be in contact with a top of the guide rail **214**, e.g., as may be seen in FIG. **11**. In such embodiments, the guide rail **214** may define a height **226** above the floor **204** of the housing **200**. Also in such embodiments a bottom surface **232**, e.g., which may be the lowermost surface of the air filter assembly **201**, may be spaced apart from the floor **204** of the housing **200** along the vertical direction V when the air filter assembly **201** is in the inserted position. For example, as illustrated in FIG. **11**, a gap **228** may thereby be defined below the air filter assembly **201**, and the gap **228** may partially define the flow path **210**, e.g., moisture may be permitted to travel along the slope of the floor **204** from the downstream side **242** of the air filter assembly **201** to or towards the upstream side **240** of the air filter assembly **201**, such as underneath the air filter assembly **201**, through the gap **228**.

(48) This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A laundry appliance, comprising: a cabinet defining an interior volume; a tub mounted within the interior volume of the cabinet; a laundry basket rotatably mounted within the tub, the laundry basket defining a chamber for the receipt of articles for treatment; a heating system in thermal communication with the chamber whereby heated air flows from the heating system to the chamber; an air filter assembly positioned between the laundry basket and the heating system upstream of the heating system, whereby a flow of return air from the chamber passes through and is filtered by an air filter of the air filter assembly before flowing to the heating system; a housing, the air filter positioned within the housing; and a drain path extending through the housing, wherein the air filter assembly is removably positioned in the housing, wherein the housing comprises a guide rail, the air filter configured to translate in a travel direction along the guide rail between an inserted position in the housing and an extracted position at least partially outside of the housing, and wherein the guide rail is discontinuous along the travel direction.
2. The laundry appliance of claim 1, wherein the air filter assembly comprises an upstream side and a downstream side, wherein the drain path extends between the upstream side of the air filter assembly and the downstream side of the air filter assembly.
3. The laundry appliance of claim 2, wherein the downstream side of the air filter assembly faces the heating system.
4. The laundry appliance of claim 2, further comprising a return duct extending from the tub to an outlet defined in the housing, wherein the upstream side of the air filter assembly faces the outlet.
5. The laundry appliance of claim 1, further comprising a return duct extending from the tub to an outlet defined in the housing, wherein the housing comprises a floor, wherein the floor of the housing defines a pitch, and wherein the pitch of the floor of housing is oriented towards the outlet.
6. The laundry appliance of claim 1, wherein the housing comprises a floor, wherein the air filter assembly comprises a guide surface, the guide surface in contact with a top of the guide rail when the air filter assembly is in the inserted position, wherein the guide rail defines a height above the

floor of the housing, and wherein a bottom surface of the air filter assembly is spaced apart from the floor of the housing along a vertical direction when the air filter assembly is in the inserted position.

7. The laundry appliance of claim 1, further comprising a water channel defined in a floor of the housing, wherein the water channel defines at least a portion of the drain path.

8. The laundry appliance of claim 7, wherein the air filter assembly is insertable into and removable from the housing through a filter gate in the housing, and wherein the water channel is defined in the floor of the housing proximate the filter gate along the travel direction.

9. A laundry appliance, comprising: a cabinet defining an interior volume; a housing mounted within the interior volume of the cabinet; an air filter assembly positioned in the housing; a drain path extending through the housing; and a return duct extending to an outlet defined in the housing, wherein the housing comprises a floor, wherein the floor of the housing defines a pitch, and wherein the pitch of the floor of housing is oriented towards the outlet.

10. The laundry appliance of claim 9, wherein the air filter assembly comprises an upstream side and a downstream side, wherein the drain path extends between the upstream side of the air filter assembly and the downstream side of the air filter assembly.

11. The laundry appliance of claim 10, wherein the downstream side of the air filter assembly faces a heating system of the laundry appliance.

12. The laundry appliance of claim 10, wherein the upstream side of the air filter assembly faces the outlet.

13. The laundry appliance of claim 9, wherein the air filter assembly is removably positioned in the housing, wherein the housing comprises a guide rail, the air filter assembly configured to translate in a travel direction along the guide rail between an inserted position in the housing and an extracted position at least partially outside of the housing, and wherein the guide rail is discontinuous along the travel direction.

14. The laundry appliance of claim 13, wherein the air filter assembly comprises a guide surface, the guide surface in contact with a top of the guide rail when the air filter is in the inserted position, wherein the guide rail defines a height above the floor of the housing, and wherein a bottom surface of the air filter assembly is spaced apart from the floor of the housing along a vertical direction when the air filter assembly is in the inserted position.

15. The laundry appliance of claim 9, further comprising a water channel defined in the floor of the housing, wherein the water channel defines at least a portion of the drain path.

16. The laundry appliance of claim 15, wherein the air filter assembly is removably positioned in the housing whereby the air filter assembly is removable from the housing by sliding in a travel direction between an inserted position in the housing and an extracted position at least partially outside of the housing, wherein the air filter assembly is insertable into and removable from the housing through a filter gate in the housing, and wherein the water channel is defined in the floor of the housing proximate the filter gate along the travel direction.

17. A laundry appliance, comprising: a cabinet defining an interior volume; a housing mounted within the interior volume of the cabinet; an air filter assembly positioned in the housing; a drain path extending through the housing; and a water channel defined in a floor of the housing, wherein the water channel defines at least a portion of the drain path, wherein the air filter assembly is removably positioned in the housing whereby the air filter assembly is removable from the housing by sliding in a travel direction between an inserted position in the housing and an extracted position at least partially outside of the housing, wherein the air filter assembly is insertable into and removable from the housing through a filter gate in the housing, and wherein the water channel is defined in the floor of the housing proximate the filter gate along the travel direction.

18. The laundry appliance of claim 17, wherein the air filter assembly comprises an upstream side and a downstream side, wherein the drain path extends between the upstream side of the air filter assembly and the downstream side of the air filter assembly.

19. The laundry appliance of claim 17, further comprising a return duct extending from the tub to an outlet defined in the housing, wherein the housing comprises a floor, wherein the floor of the housing defines a pitch, and wherein the pitch of the floor of housing is oriented towards the outlet.
20. The laundry appliance of claim 17, wherein the housing comprises a guide rail, wherein the air filter assembly comprises a guide surface, the guide surface in contact with a top of the guide rail when the air filter assembly is in the inserted position, wherein the guide rail defines a height above the floor of the housing, and wherein a bottom surface of the air filter assembly is spaced apart from the floor of the housing along a vertical direction when the air filter assembly is in the inserted position.
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